Alberto Colombo

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Education

2017 – (July 2020) **M.Sc. Politecnico di Milano, IT** in Mathematical Engineering
Avg grade: 28.86/30
Courses: Real and functional analysis, Algorithms and parallel computing, Digital signal processing, Stochastic dynamical models, Advanced partial differential equations, Calculus of variations, Applied statistics, Model identification and data anlysis, Bayesian statistics, Electronics, Plasma physics, Advanced programming for scientific computing.

2015 – 2017 **Bachelor Politecnico di Milano, IT** in Mathematical Engineering
Final grade: 110/110 cum laude
Courses: Calculus I/II/III, Linear algebra, Numerical analysis, Experimental physics I/II, Probability, Statistics I/II, Operational research, Automation and control, Economics, Chemistry, Informatics, Theoretical and applied mechanics, Thermodynamics, Partial differential equations. **Additional courses (extra credits):** Quantum physics, Data bases.

2009 – 2014 **High school - ISISS L.Geymonat - Tradate, IT**
Final grade: 100/100

Employment History

Aug. 2019 – now **Math Tutor** English Business Services, Milano - IT
Part-time trainer for the quantitative reasoning section of different graduate admission tests (GMAT, GRE, SAT).

Nov. 2018 – May 2019 **Data Science Intern** Campari S.p.A., Sesto S.Giovanni, Milano - IT
Intern in the new-born advanced analytics division. Solving business problem through data analysis and statistical modeling (demand forecasting, customer classification and profiling, autonomous CV screening, optimal sales force allocation). Communicating results to non-technical marketing/business executive decision makers.

July 2014 **Flight simulator intern** Leonardo Helicopters, Sesto Calende - IT
Short internship in the flight simulator software team. The internship was offered to the top 5 partecipants of "Valerio Filippini" physics prize.

Projects

ongoing **M.Sc. thesis** – Supervisor: U.Spagnolini
Title: *Multiple sensors augmented positioning for high performance vehicles*
Best Presentation Award at the 2019 Huawei Italy University Challenge (2° place)
Development of an accurate positioning system via bayesian sensor fusion. Application of the resulting method on high performance vehicles (i.e. F1 racing cars).

ongoing **Advanced programming for scientific computing** – Supervisors: L.M.Sangalli, L.Formaggia
Development of the fdaPDE Cran package (R software) in C++, concerning spatial regression with partial differential equation regularization.
The project is carried on by different groups of students of different years. Specifically, my case is about generalized linear models on 2D surfaces.

Projects (continued)

2019 **Bayesian Statistics project** – Supervisors: A.Cadonna, A.Guglielmi
Statistical analysis of spatio-temporal data (telephone traffic) on the area of Milan. Simulation via MCMC for ANOVA-DDP with sparse CAR prior.

2018 **Applied Statistics project** – Supervisors: A.Menafoglio, P.Secchi
Statistical analysis of Mappa dei rischi dei comuni italiani, particularly devoted to migration aspects within Italy. The dataset was provided by the Casa Italia department of Italian Government.

2017 **Bachelor thesis** – Supervisor: E. Lanzarone
Title: *Analysis of the predictive capabilities of electric current and pressure on the output of electrospinning process*
Developed in collaboration with IMATI division of CNR in the framework of project nano-TWICE.

Skills

Languages
- **Italian:** Native speaker.
- **English:** Strong reading, writing and speaking competencies. (TOEIC 2017: 845/990)

Coding
- **R:** Expert – dplyr, shiny, ggplot2, stan.
- **C++:** Highly proficient – eigen, expression templates, design pattern, MPI
- **MATLAB:** Higly proficient – linear algebra, differential equations, signal processing, simulink.
- **Python:** proficient – numpy, pandas, spark, matplotlib, sklearn.
- **C, Pascal:** basic experience.

Other softwares
- LaTeX, SQL, Linux(Ubuntu), Windows, Office.

Misc.
- analytical modeling, numerical methods for pde, teaching, training, teamworking, personal communication, public speaking.

Personal Experience

Personal Study

ongoing **NLP with Deep Learning**: public Stanford course CS224N held by Christopher Manning (assignments developed so far on my GitHub).

2020 **The Book of Why**: introduction to causal graphs by Judea Pearl.

2018 **Crash Course in Data Science**: Series of seminars on Data Science topics offered in 2018 in the passion-in-action initiative at DEIB. For a 2019 reference of the same course: Crash Course in Data Science – Passion in Action.

Theatre Experience

2017 – now **Executive Officer** – Cultural association Teatro delle Biglie
Organization and management (lectures, courses, shows,..).

2019 **Assistant Teacher** – Cultural association Teatro delle Biglie
Teacher in a four months class (15 people) for performing actors.

2019 **Performer** – Gelmeroda VIII
Stage actor for an independent play held in Politecnico di Milano's auditorium.

About

Accurate, precise and methodological, can't rest until work is excellently done. My extreme curiosity led me to be a very passionate learner in many different fields, from science to technology and history. I feel comfortable working in teams and doing public speeches.

Career Aspirations

I am strongly interested in Machine Learning and specifically its application to Natural Language Processing. In future years I plan to start a PhD and, more in general, my long term goal is to become a researcher and a professor in this field. For these reasons I am currently looking for research opportunities involving NLP.

Hobbies & Interests

Board-games, analog photography, hiking, biking, theatre, cognitive sciences, evolutionary biology, ancient history.